An unsigned underflow in MsQuic's CUBIC

Evidence report: the bug, a two-line fix, and why it is not our mistake

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Abstract

In MsQuic's CUBIC congestion controller, the Reno-friendly window `AimdWindow` is grown *before* the accumulator `AimdAccumulator` is reduced by it. Both are `uint32`, so whenever an ACK carries the accumulator past the window by less than one packet, the subtraction wraps to about 4.29×10^9. From then on every ACK grows `AimdWindow` (and, in the Reno-friendly region, the congestion window) by one packet: CUBIC behaves like slow start in congestion avoidance. Swapping the two lines fixes it. This report collects the evidence that the bug is real, is in *unmodified upstream* MsQuic (`main` at `3c9ec01c1`, 10 November 2026), is caused by these two lines alone, and that the fix changes nothing else. The unit-test and debugger evidence was also re-run independently on GitHub's servers, against upstream `main` cloned directly from GitHub. On an emulated network path, logging the accumulator shows a wrap in all 20 upstream runs and in none of the 20 fixed runs. In plain bulk transfers without a rate-limited phase the wrap is usually undone by the next ACK, but it persisted in 10 of 40 upstream runs (0 of 40 fixed), and those runs lost 5–130 times more packets than the fixed build's median. Every claim below is followed by raw output of a tool (test runner, debugger, trace) that anyone can regenerate with the commands in Section 9.

1 The bug and the fix

`src/core/cubic.c`, `CubicCongestionControlOnDataAcknowledged` (upstream `main`, lines 659–662)

```
if (Cubic>AimdAccumulator > Cubic->AimdWindow) {
   Cubic->AimdWindow += DatagramPayloadLength;
    Cubic>AimdAccumulator -= Cubic->AimdWindow;
   Cubic->AimdWindow += DatagramPayloadLength;
}
```

What the code is meant to do. The comment above it says: record ACKed bytes in an accumulator "until at least a window (or two windows) worth of bytes are ACKed, and then increase the window by 1 MTU". So when the accumulator passes the window, one window's worth of bytes should be taken out of it and the window grown by one packet.

What it does. It grows the window first and then subtracts the *grown* window, which is one packet more than was accumulated. A worked example with the numbers caught by the debugger in Section 3:

	upstream order	fixed order
`AimdWindow` before	11 211	11 211
`AimdAccumulator` after this ACK	11 704	11 704
`AimdAccumulator` after subtracting	$11\,704 - 12\,443 \rightarrow$ **4 294 966 557**	$11\,704 - 11\,211 =$ **493**
`AimdWindow` after	12 443	12 443

With the wrapped value the condition `AimdAccumulator > AimdWindow` is true on every following ACK, and each of them adds one packet to `AimdWindow`. Because CUBIC sets `CongestionWindow = AimdWindow` whenever `AimdWindow` exceeds the cubic curve, the congestion window grows by one packet per ACK, i.e. it doubles every round trip, until the wrapped value has been counted down. Congestion events reset `AimdWindow` but not the accumulator, so the effect survives losses.

Why it went unnoticed. Three reasons, each shown below. (1) If the window is an exact multiple of the packet size and every ACK adds the same amount, the next ACK wraps the accumulator straight back to a small value and nothing visible happens; upstream's tests use such round numbers. (2) Two existing upstream tests *do* trigger the wrap, but neither checks the accumulator, so they pass (Section 3). (3) Upstream's test `AIMD_AccumulatorTriggersWindowGrowth` chooses its ACK size, in its own words, "to avoid unsigned underflow in the subtraction (AimdAccumulator -= AimdWindow)".

How long it has been there. `git blame` on upstream `main` attributes the condition and the subtraction to the CUBIC rewrite in PR #2269 (`586141e1a`, January 2022), which already grew the window first (by `Paths[0].Mtu` at the time). PR #2296 (`ba0819664`) changed only the amount added, to `DatagramPayloadLength`. The order is therefore in every release since v2.0.0; it is not a recent regression.

2 Evidence 1: reproducers on unmodified upstream `main`

We checked out upstream `main` (`3c9ec01c1`, 10 Nov 2026) in a separate worktree, with none of our other changes, and added two unit tests to upstream's own `CubicTest.cpp` using upstream's own test fixture:

- `AIMD_AccumulatorNoUnderflow` (state level): sets up an accumulator 50 B below the window and acknowledges 100 B. The accumulator must then hold 50.
- `AIMD_PerPacketAcksGrowAtMostOneMssPerWindow` (behaviour level): *no internal field is set.* After one loss, the test sends a full window per round trip and acknowledges every packet separately, as a real path does, using only the public callbacks. It asserts what the algorithm promises: `AimdWindow` grows by at most one packet per `AimdWindow` bytes acknowledged.

Before the fix (upstream `cubic.c` unchanged), verbatim:

```
src/core/unittest/CubicTest.cpp:2342: Failure
Expected equality of these values:
  Cubic->AimdAccumulator
    Which is: 4294966114
  50u
    Which is: 50
[  FAILED  ] CubicTest.AIMD_AccumulatorNoUnderflow (0 ms)
round 0: cwnd 11211 -> 11957, AimdWindow 11211 -> 11211, AimdAccumulator 5544
round 1: cwnd 11957 -> 12384, AimdWindow 11211 -> 11211, AimdAccumulator 11088
round 2: cwnd 12384 -> 23531, AimdWindow 11211 -> 23531, AimdAccumulator 4294808370
src/core/unittest/CubicTest.cpp:2388: Failure
Expected: (Cubic->AimdWindow - AimdBefore) <= (MaxGrowth), actual: 12320 vs 2464
[  FAILED  ] CubicTest.AIMD_PerPacketAcksGrowAtMostOneMssPerWindow (0 ms)
```

In round 2 the congestion window goes from 12 384 to 23 531 bytes in one round trip of congestion avoidance (+90 %), and `AimdWindow` grows by ten packets where two is the maximum.

After swapping the two lines (nothing else in `cubic.c` changed):

```
[       OK ] CubicTest.AIMD_AccumulatorNoUnderflow (0 ms)
round 0: cwnd 11211 -> 11957, AimdWindow 11211 -> 11211, AimdAccumulator 5544
round 1: cwnd 11957 -> 12384, AimdWindow 11211 -> 11211, AimdAccumulator 11088
```

```
round 2: cwnd 12384 -> 12716, AimdWindow 11211 -> 12443, AimdAccumulator 6037
round 3: cwnd 12716 -> 12985, AimdWindow 12443 -> 12443, AimdAccumulator 12197
round 4: cwnd 12985 -> 13675, AimdWindow 12443 -> 13675, AimdAccumulator 5914
round 5: cwnd 13675 -> 13675, AimdWindow 13675 -> 13675, AimdAccumulator 12690
round 6: cwnd 13675 -> 14907, AimdWindow 13675 -> 14907, AimdAccumulator 5791
round 7: cwnd 14907 -> 14907, AimdWindow 14907 -> 14907, AimdAccumulator 13183
[       OK ] CubicTest.AIMD_PerPacketAcksGrowAtMostOneMssPerWindow (0 ms)
```

Rounds 0–1 are identical in both runs: the two versions only differ from the moment the accumulator crosses the window.

3 Evidence 2: a debugger catches the wrap in upstream code

To take our own code out of the picture entirely, we let `gdb` watch upstream's unmodified `cubic.c`. A breakpoint on line 661 (`AimdAccumulator -= AimdWindow`) fires only if the subtraction is about to go below zero, prints both operands and the `uint32` result, and records which test is running. Run over *upstream's own, unmodified* `CubicTest` suite:

```
[ RUN      ] CubicTest.CongestionAvoidance_BoundedGrowthClamp
=== underflow 1 at cubic.c:661  AimdAccumulator -= AimdWindow ===
AimdWindow (already grown on line 660) = 1232
AimdAccumulator before the subtraction = 1200
AimdAccumulator after (uint32 result)  = 4294967264
#1  CubicTest_CongestionAvoidance_BoundedGrowthClamp_Test::TestBody at src/core/unittest/CubicTest.
    cpp:761
[       OK ] CubicTest.CongestionAvoidance_BoundedGrowthClamp (1 ms)
...
[ RUN      ] CubicTest.CongestionAvoidance_TimeGapOverflowProtection
=== underflow 2 at cubic.c:661  AimdAccumulator -= AimdWindow ===
AimdWindow (already grown on line 660) = 1232
AimdAccumulator before the subtraction = 1200
AimdAccumulator after (uint32 result)  = 4294967264
#1  CubicTest_CongestionAvoidance_TimeGapOverflowProtection_Test::TestBody at src/core/unittest/
    CubicTest.cpp:1934
[       OK ] CubicTest.CongestionAvoidance_TimeGapOverflowProtection (1 ms)
[  PASSED  ] 52 tests.
Total underflows: 2
```

Upstream's tests reach the bug twice and pass anyway. The same breakpoint on the behavioural reproducer shows the wrap that causes the jump in round 2: `AimdWindow = 12443`, `AimdAccumulator = 11704`, result `4294966557`, `CongestionWindow = 12384`.

4 Evidence 3: the fixed behaviour is the specified behaviour

The fix is not merely different; it matches what the code comment and the CUBIC specification require.

- MsQuic's comment: the AIMD window uses "a slope of 0.5MSS/RTT ... until AimdWindow reaches WindowPrior, and then the slope is increased to 1MSS/RTT". RFC 9438 Section 4.3 gives the same Reno-friendly slope, $\alpha_{cubic} \approx 0.53$ segments per RTT for $\beta = 0.7$.
- With the fix, the reproducer's `AimdWindow` grows by one packet every two round trips (11 211, 12 443, 13 675, 14 907 at rounds 1, 2, 4, 6): **0.5 MSS/RTT**, exactly the specified slope.
- Upstream grows `AimdWindow` by 10 packets in one round trip, $\approx 20\times$ the specified slope, and `CongestionWindow` with it.

5 Evidence 4: deterministic harness, accumulator printed

Scenario F drives MsQuic's `cubic.c` through a fixed sequence: a loss, a long rate-limited period, the application resuming, a second loss, then one ACK per packet with the window fully used. Two builds whose `cubic.c` differ only in the two lines:

Round after	upstream order		fixed order	
the loss	`AimdAccumulator`	cwnd (KB)	`AimdAccumulator`	cwnd (KB)
0	4 293 692 898	89	28 690	70
1	4 284 099 738	178	63 186	71
2	4 245 760 746	355	28 559	72
3	4 092 442 506	710	64 287	73
4	3 479 245 002	1 419	29 660	74
5	1 026 605 898	2 839	66 620	75

Table 1: Scenario F. With the upstream order the accumulator holds a wrapped value near 2^{32} that every ACK reduces by about one window, the window grows on every ACK and cwnd doubles every round trip; with the fix it stays below the window and cwnd grows by about one packet per round trip.

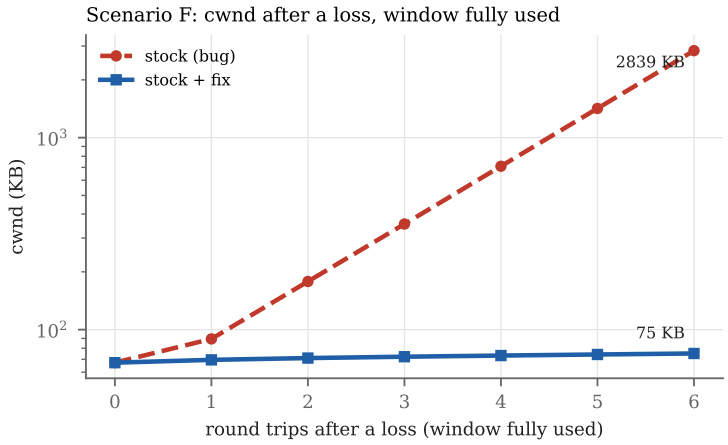


Figure 1: Scenario F, cwnd per round after the loss (log scale).

6 Evidence 5: effect on a real network path

Isolation. These runs use MsQuic `464c873c0` (31 August 2026, the base of our project), not today's `main`. Between the two, `cubic.c` changed only in how the active path is looked up (`Connection->Paths[0]` became `QuicPathGetActive(...)`, 12 places, plus line wrapping); the AIMD code and all window arithmetic are identical. The "upstream" build is that commit's CUBIC. The "fixed" build is our ablation source with every one of our rate-limited-sender changes compiled out by macros, plus the two-line swap. We checked that this configuration is exactly equivalent: in our deterministic harness (504 trace rows) it produces the same trace as upstream plus the swap, and with the swap undone the same trace as upstream, so the only behavioural difference between the two builds is the swap. The direct-observation run below goes one step further: its two builds are compiled from sources that differ textually in the two lines only (`experiments/acc_proof/S0_vs_S1.diff`).

Testbed. Linux network namespaces with a `netem` bottleneck (10–50 Mbit/s, 41–81 ms RTT, 100-packet drop-tail queue, segmentation offloads off). A QUIC sender sends 3 MB, then 6000 B every

10 ms, then 8 MB. 10 runs per build and setting, build order rotated; medians, two-sided permutation tests (20 000 permutations).

	full-packet CA increases		cwnd at resume (KB)		lost after resume (KB)	
Setting	upstream	fixed	upstream	fixed	upstream	fixed
6 s rate-limited, 10 Mbit/s	33 %	1.0 %	668	229	3413	82
28 s rate-limited, 10 Mbit/s	23 %	0.7 %	670	671	1094	1094
6 s, RTT 81 ms	25 %	0.2 %	762	232	2800	84
10 s, 50 Mbit/s	22 %	0.4 %	668	288	3002	140
25 Mbit/s + TCP flow	11 %	5.7 %	128	88	106	47

Table 2: Upstream vs. upstream + the two-line fix on a real (emulated) network path; medians of 10 runs. The signature of the bug (a congestion-avoidance increase of exactly one packet on a single ACK) drops from 11–33 % to 0.2–6 % of increases ($p \leq 0.012$ in every setting). Where the bug has time to act (6–10 s) it inflates the window 2.3–3.3× and the loss after resumption 21–42× ($p < 0.004$). After 28 s plain CUBIC reaches the same limit on its own, so the window and loss are equal ($p > 0.6$): the fix does not change anything it should not.

Direct observation of the accumulator. We repeated two settings with builds that log `AimdWindow` and `AimdAccumulator` on every ACK. The two sources differ textually only in the swap (`experiments/acc_proof/S0_vs_S1.diff`) and contain none of our rate-limited-sender code. Both are upstream `464c873c0`'s `cubic.c` plus a trace logger, compiled in the same tree. That tree also holds our test client (`sample.c`) and one extra field in `cubic.h` that neither source uses; both are the same for the two builds. Since `AimdAccumulator` only ever holds acknowledged bytes (at most a few hundred KB here), a value $\geq 2^{31}$ can only be a wrap; no model or threshold tuning is involved.

Setting	Build	runs with a wrap	wrap events (median)	full-packet CA incr.	cwnd at resume (KB)	lost after resume (KB)	max `AimdAccumulator` (bytes)
6 s RL, 10 Mbit/s	upstream	10/10	22	32 %	668	3351	4.29×10^9
	fixed	0/10	0	0.9 %	218	72	163 285
25 Mbit/s + TCP	upstream	10/10	84	16 %	146	765	4.29×10^9
	fixed	0/10	0	3.6 %	80	32	129 874

Table 3: Accumulator logged on every ACK, 10 runs per build and setting (medians; RL = rate-limited; "max `AimdAccumulator`" is the largest value seen in any run of the setting). Upstream vs. fixed: $p = 0.0001$ for cwnd, loss and full-packet share in the 6 s setting; $p = 0.016$, 0.008 and 0.004 in the TCP setting. The 6 s row reproduces the four-build campaign above (668 vs. 229 KB, 3413 vs. 82 KB).

Across all ACKs the upstream build processed 90 764 ACKs, and after 27 940 of them `AimdAccumulator` held a wrapped value; the fixed build processed 92 400 ACKs and never did. In 522 fixed-build ACKs `AimdAccumulator` was larger than `AimdWindow` without being wrapped. We checked each one: all 28 such stretches start exactly at a congestion event, where MsQuic lowers `AimdWindow` but keeps `AimdAccumulator`, and `AimdAccumulator` stays unchanged through recovery until the next growing ACK subtracts the window normally. This is upstream behaviour unrelated to the swap. Figure 2 shows one representative run per build (the run with the median number of wrap events): the upstream accumulator sits at $\approx 2^{32}$ or jumps there dozens of times, and each time cwnd climbs by one packet per ACK; the fixed accumulator never exceeds 1.7×10^5.

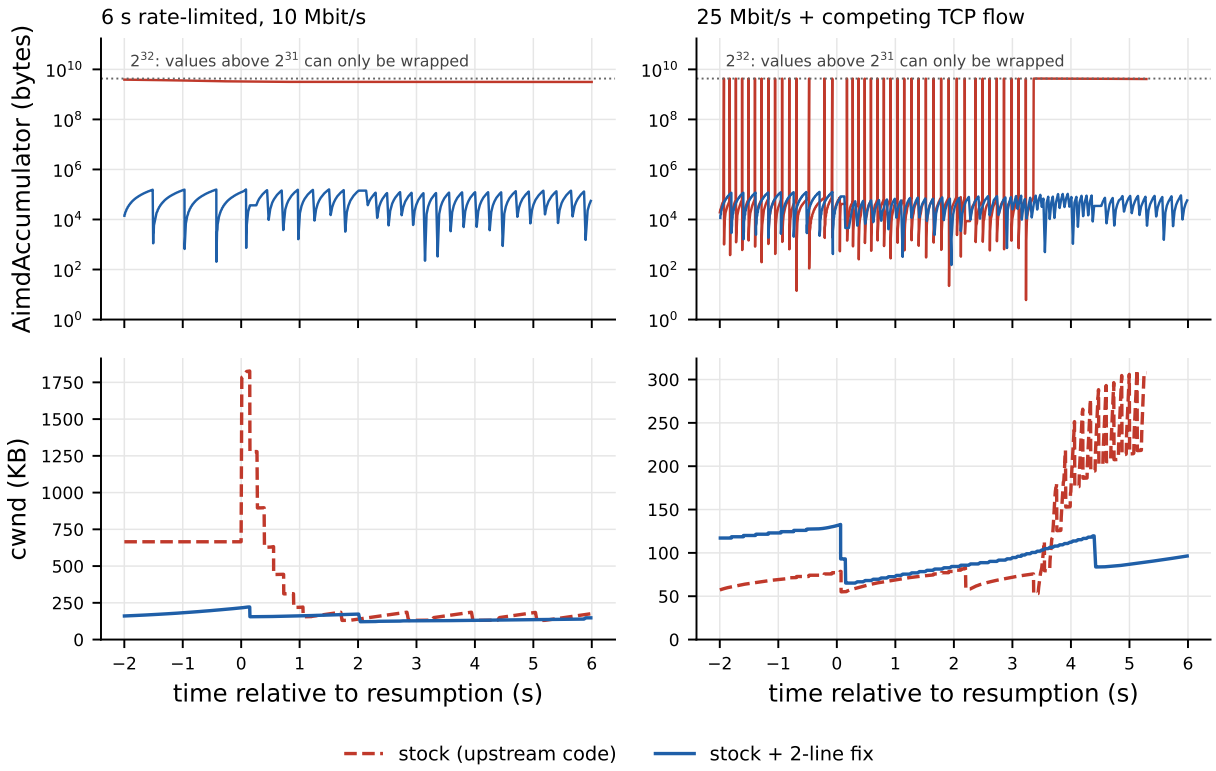


Figure 2: `AimdAccumulator` (log scale, top) and cwnd (bottom) on every ACK around the resumption of bulk sending, upstream vs. upstream + the two-line fix. Left: 6 s rate-limited at 10 Mbit/s (the upstream accumulator wrapped during the rate-limited phase and stays wrapped, so cwnd is already 668 KB at resumption). Right: 25 Mbit/s with a competing TCP flow.

Plain bulk transfers. Our largest effects come after a rate-limited phase, the scenario of our project. To see whether the bug matters without one, we ran the same two logging builds on single bulk transfers with no rate-limited phase (10 runs per build and setting, build order rotated): 30 MB at 10 Mbit/s with 40 and 80 ms RTT, 100 MB at 50 Mbit/s, and 60 MB at 25 Mbit/s shared with a TCP CUBIC flow (`sudo experiments/run_campaign.sh bulk`).

The wrap happened in all 40 upstream runs and in none of the 40 fixed runs (30–515 wraps per upstream run), but in most runs it did no harm. A wrapped accumulator is undone by the next ACK whenever that ACK adds more than the shortfall, which is always less than one packet: the `uint32` addition wraps it back to a small value. With the usual ACK of two packets this always happens. The wrap *persists* only if the next ACK adds less than the shortfall. That happened in 10 of the 40 upstream runs, and in all 10 the cause was the same: a single-packet ACK (1472 B) while `AimdWindow` was below `WindowPrior`, where the code counts only half of each ACK (736 B). From then on every ACK subtracts the growing window from a value near 2^{32}, so the accumulator stays wrapped for thousands of ACKs (1 248 to 20 464 here), usually until the end of the transfer.

Setting	upstream: wrap persisted	upstream: wrap undone	fixed	runs with persistent wrap (upstream vs. fixed)
10 Mbit/s, 40 ms	—	164–180 (10 runs)	171–182	0/10 vs. 0/10
10 Mbit/s, 80 ms	**1 056–3 241** (3)	192–202 (7)	194–203	3/10 vs. 0/10
50 Mbit/s, 40 ms	**582–6 244** (2)	114–131 (8)	114–123	2/10 vs. 0/10
25 Mbit/s + TCP	**845–10 935** (5)	65–98 (5)	73–100	5/10 vs. 0/10

Table 4: Lost packets per bulk transfer (range over runs; number of runs in brackets). Last column: runs in which the wrap persisted for more than 100 ACKs, upstream vs. fixed (10/40 vs. 0/40 overall, Fisher exact test $p = 0.001$). Upstream runs whose wraps were all undone are indistinguishable from the fixed build in every metric we measured; runs with a persistent wrap lose 5–130× more packets than the fixed build's median.

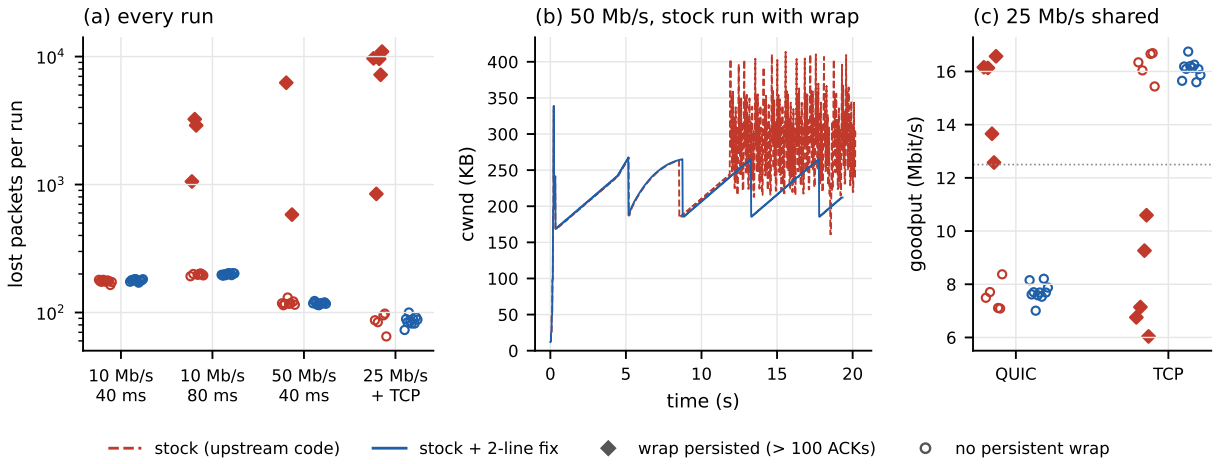


Figure 3: Plain bulk transfers. (a) Lost packets in every run; filled diamonds are upstream runs in which the wrap persisted. (b) cwnd at 50 Mbit/s in the upstream run with the longest wrap and a typical fixed run: once the wrap persists (at about 12 s), cwnd grows by one packet per ACK until a loss, over and over. (c) Goodput of the QUIC transfer and the competing TCP flow at 25 Mbit/s, per run; the dotted line is an equal share.

Without competition the link is full either way, so completion time does not change (medians within 0.01 s); what a persistent wrap adds is self-inflicted loss and queue overflow at the bottleneck. With a competing TCP flow it also changes the split of the link. In the five upstream runs with a persistent wrap the QUIC transfer got a median 16.1 Mbit/s and TCP 7.1 Mbit/s; in the fixed runs QUIC got 7.7 and TCP 16.1 Mbit/s. We note honestly that with the fix the QUIC flow gets *less* than half of the link against Linux TCP CUBIC. The bug happens to push it the other way, but by making it far more aggressive than RFC 9438 allows (one packet per ACK instead of 0.5 packet per RTT), not by correcting whatever causes that gap, which we have not investigated.

7 Evidence 6: nothing else changes

Upstream `main` (`3c9ec01c1`)	Tests	Passed	Skipped	Failed
unmodified	563	559	4	0
+ fix, 2 new tests, 2 updated expectations	565	561	4	0

Table 5: MsQuic's full core test suite (`msquiccoretest`, all 39 suites).

The only difference is the two new tests. The fix touches only the branch taken when the accumulator crosses the window; whenever it crosses by a full packet or more, the old and new order give the

same `AimdWindow` and differ only in the accumulator's remainder (which the old order understated by one packet).

The two updated expectations. Two upstream tests hard-code the old order: `AIMD_AccumulatorTriggersWindowGrowth` expects $1432 - (100 + 1232) = 100$, i.e. "subtract the grown window", after choosing its ACK size "to avoid unsigned underflow"; and `AIMD_SequentialLinearConvergence` computes its expected values in the old order. Both now expect "subtract the window that was crossed, then grow" (the accumulator keeps $1432 - 100 = 1332$ bytes). No other test changed.

8 How we ruled out mistakes on our side

- **Our patch is not involved.** Evidence 1, 2 and 6 use a fresh checkout of upstream `main` with nothing of ours except the two tests. In Evidence 5 our patch is compiled out (four-build campaign) or absent (direct-observation run).
- **Exactly two lines.** In Evidence 1, 2, 4 and the direct-observation run the compared sources differ textually only in the swap; the diffs are in `experiments/acc_proof/` and the source hashes are recorded.
- **Independent tools.** The wrap is reported by gdb reading upstream's variables, and by MsQuic's own test runner; our analysis scripts are not needed for Evidence 1–4.
- **Run by a third party.** Evidence 1, 2 and 6 were re-run on GitHub's servers by a GitHub Actions workflow (`.github/workflows/aimd-proof.yml`). It clones `microsoft/msquic` directly from GitHub, at `3c9ec01c1` and at the then-current `main`, and checks every expected outcome itself: both reproducers fail, gdb catches the wrap in upstream's own tests, the change to `cubic.c` is exactly one moved line, and with it the reproducers, all CUBIC tests and the full core suite pass. Both jobs succeeded on 12 November 2026; the log and raw outputs are hosted by GitHub, not by us: `https://github.com/Abhirams144/msquic-rate_limited_sender_increase/actions/runs/43286985783`.
- **No inference needed.** In the direct-observation run the wrap is read from the logged variable itself ($\geq 2^{31}$), not inferred from cwnd: 20/20 upstream runs vs. 0/20 fixed runs.
- **Natural inputs.** The behavioural reproducer and the network runs set no internal state; the bug appears on its own.
- **When the bug is harmless.** A wrap is undone by the next ACK whenever that ACK adds more than the shortfall (less than one packet), which is the normal case in steady bulk transfer. It persists after a single-packet ACK while `AimdAccumulator` counts half of each ACK (below `WindowPrior`), which is common when sending is rate-limited or another flow shares the queue, and happened in 10 of 40 plain bulk transfers (Section 6). The effect is therefore all-or-nothing per occurrence, not a small bias.

9 Reproducing everything

```
# 1. upstream main + the two tests, before and after the fix
git -C msquic worktree add --detach ../upstream-main 3c9ec01c1
cd upstream-main && git submodule update --init submodules/{clog,quictls,googletest}
cmake -B build-test -DQUIC_BUILD_TEST=ON -DQUIC_TLS_LIB=quictls -DCMAKE_BUILD_TYPE=Debug
cat ../experiments/acc_proof/reproducer_tests.cpp >> src/core/unittest/CubicTest.cpp
cmake --build build-test --target msquiccoretest
build-test/bin/Debug/msquiccoretest --gtest_filter='CubicTest.AIMD_*'      # fails
git apply ../experiments/acc_proof/fix_on_upstream_main.diff               # the 2 lines
# 2. catch the wrap with gdb (on the unfixed build)
gdb -batch -x ../experiments/acc_proof/catch_wrap.gdb \
```

```
    --args build-test/bin/Debug/msquiccoretest --gtest_filter='CubicTest.*'
# 3. full core suite: build-test/bin/Debug/msquiccoretest --gtest_brief=1
# 4. network: experiments/acc_proof/build_acc.sh; sudo experiments/run_campaign.sh acc
#             python3 experiments/acc_proof/analyze_acc.py
#    bulk:    sudo experiments/run_campaign.sh bulk
#             python3 experiments/acc_proof/analyze_bulk.py
```